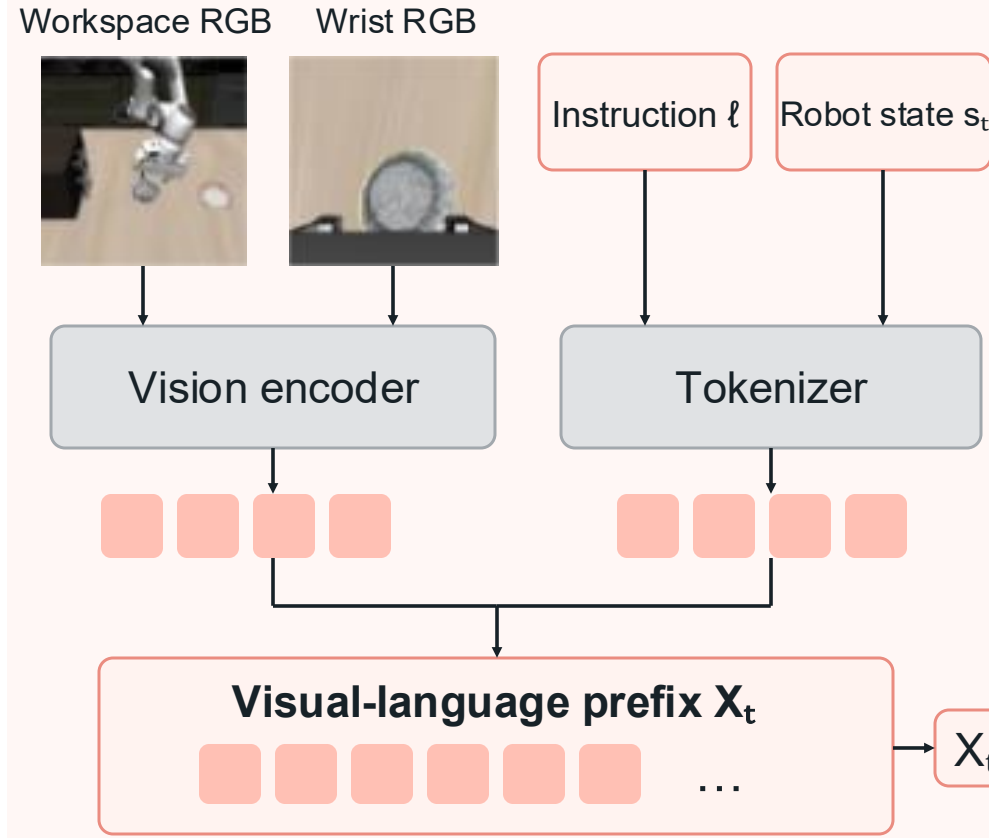
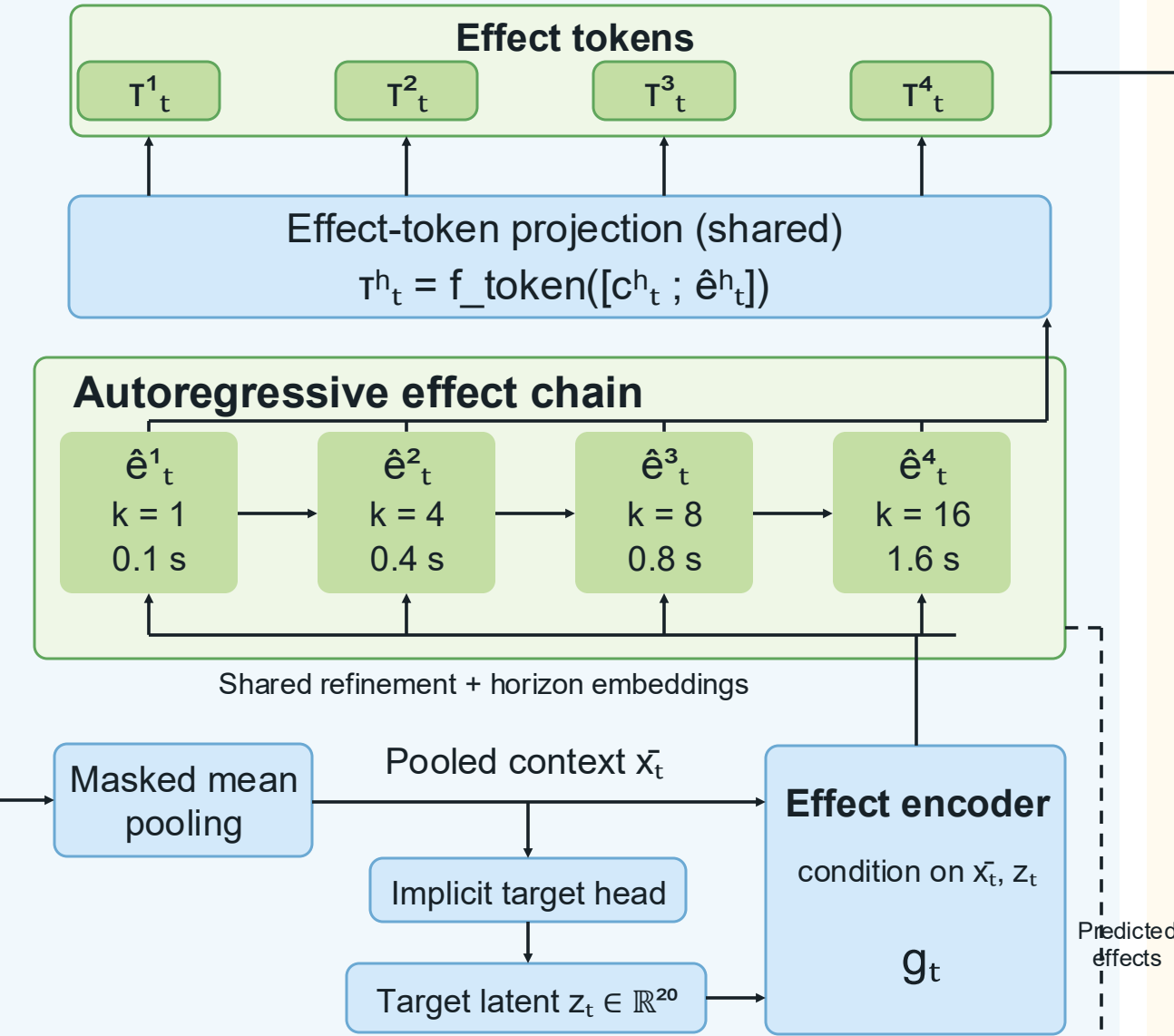


ArTE-VLA: Autoregressive Target Effects for Vision-Language-Action Models

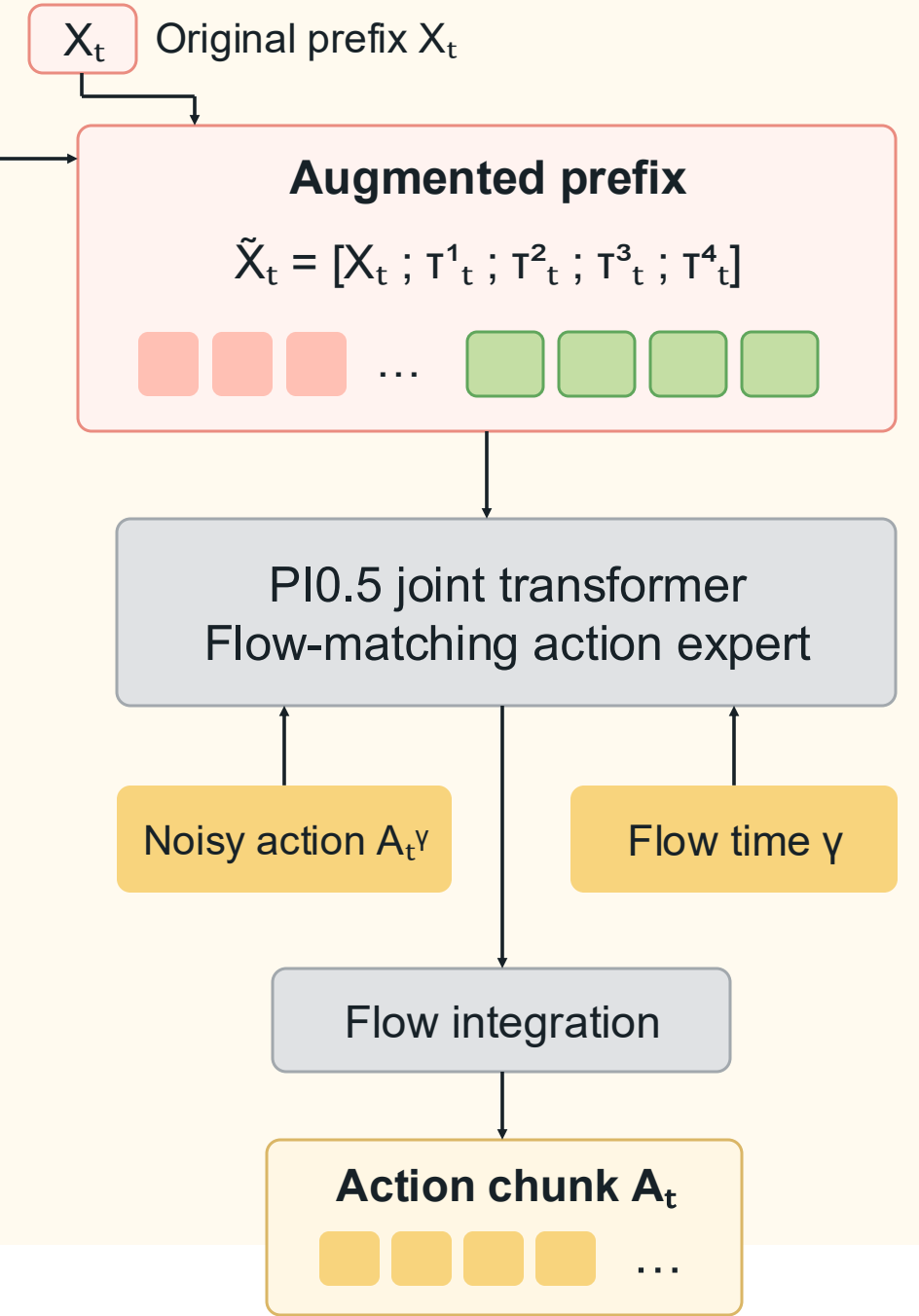
(a) Multimodal context



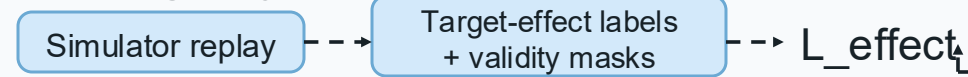
(b) Autoregressive target effects



(c) Effect-conditioned action generation



Training only



13-D effect: Δp (3) translation Δr (3) rotation v (3) linear velocity ω (3) angular velocity q (1) activity

Smooth-L1 + activity BCE

■ Multimodal tokens

■ Predicted effect tokens

■ Action tokens

→ Forward path

- - - - - Training supervision

$$L = L_action + \lambda_E L_effect$$